Perpendicular bisector of [IIA]

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Let ABC be a triangle. Let I and I_A be the centres of the incircle of ABC and of the excircle opposite A. The perpendicular bisector of $[II_A]$ meets (AB) (resp. (AC)) at L (resp. N). Show that if (LN) is tangent to the circumcircle of ABC, then ABC is isosceles.